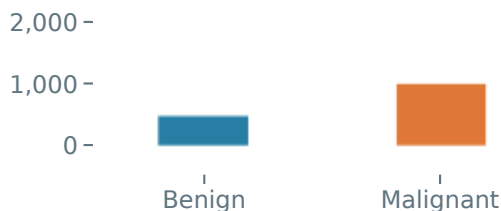


Start with the question, then test what the chart preserves

Example: How do two groups differ across their distributions?

Compare the 10th, 25th, 50th, 75th and 90th percentiles.

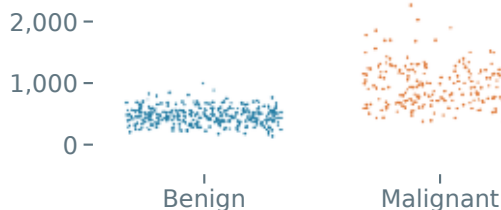


A Group averages

Shows one average per group.

The five requested percentiles are not encoded.

Recovery error: 0.6621

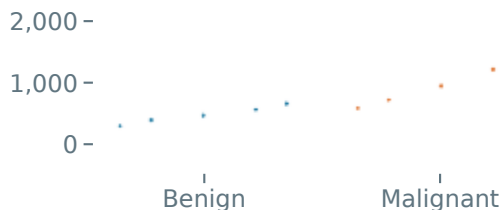


B Individual observations

Shows the observed values.

A pixel-reading algorithm estimates the five percentiles.

Recovery error: 0.0236



C Five percentile markers

Shows the five requested values.

A strong comparison chart: it directly encodes the information being tested.

Recovery error: 0.0110

Lower error means a more accurate numerical answer from the pixels.

Next test: choose at one display size, freeze the choice, and compare it with a strong default at a different display size. See Figure 3.